*Reclassification footnote (2026-05-26 EOD addendum, binding 2026-05-26)* This paper was drafted prior to the 2026-05-26 C1/C2/C3 epistemic-category reclassification (Math411-AddB §10 operator binding). Post-reclassification, the canonical TECT positioning is: *TECT is an emergent vacuum-and-gravity framework (7 C1 pillars at T6/T7) with an unresolved gauge-sector completion (Pillar 6 standard routes EXHAUSTED) and phenomenological dark-matter/quantum-origin pillars open.* The TOE-level claim is SUSPENDED until a non-standard rescue is independently derived. Pillar 6 closure programme: all four standard promotion routes (Pathway B SO(4) cubic-irrelevance, $T_{2u}$ doubling, Stueckelberg-direct, Pathway A direct SO(10)) are REFUTED or STRUCTURALLY INSUFFICIENT (Math410, Math410-AddA, Math411, Math411-AddA, Math411-AddB). Pillar 11.B is DOUBLY BLOCKED at TECT-natural (Math412-AddA, Math412-AddB). Any wording in this paper stronger than its present tier supports per `Docs/status/TOE-FACT-SHEET.md` should be re-calibrated against Math411-AddB §10.

# A conditional Brazovskii-universality programme for the body-centred-cubic topological condensate: five axioms, five obstruction bounds, and a non-rigorous-class roadmap

Jusang Lee*

*Independent researcher, Republic of Korea*

(Dated: May 27, 2026)

We argue that the minimal microscopic TECT core lies in the Brazovskii-type shell-forming class, and we formulate a conditional Brazovskii-universality programme for the body-centred-cubic (BCC) topological condensate of the Topological Energy Condensate Theory (TECT). Five structural axioms (C1–C5) define the candidate class membership, and five obstruction bounds (O1–O5) are stated as conditions that must be verified for the candidate membership to be upgraded to a theorem. Numerical estimates derived from harmonic-oscillator barrier-height analysis are consistent with O1–O5 in current measurements; we deliberately label this evidence as "conditional consistency", not "rigorous closure", because the estimates rely on branch-continuation numerics and not on unbiased spontaneous-emergence simulations. Full Brazovskii-class certification remains *open* pending (i) the unbiased emergence test of the shell-forming instability and (ii) the continuum-limit RG-flow estimate beyond two loops. The conditional programme nevertheless provides a structured roadmap for tying TECT to the canonical Brazovskii literature and for organising the falsification gates that must be cleared before mean-field critical-exponent statements can be adopted as TECT predictions.

## I. INTRODUCTION

The universality class of a phase transition determines the critical exponents and scaling functions that govern the system's approach to criticality. For the Brazovskii class, first identified in superfluid helium [1], the key feature is the presence of a finite-wavelength modulation in the free-energy landscape, parametrized by the wavenumber $q_0$. This leads to a lower critical dimension $d_c = 1$ and a reentrant phase diagram with a sponge-like morphology in certain parameter regimes.

The TECT framework postulates a BCC condensate governed by a Brazovskii-type free energy [2]. However, the *topological* aspect of the TECT setting — the appearance of $O(n)$ order parameters on a discrete lattice, the coupling to emergent gravity, and the freeze-out scenario — introduces additional constraints not present in the canonical laboratory Brazovskii systems. The question arises: does TECT's BCC condensate rigorously belong to the Brazovskii class, or does the topological context force a different effective class?

This paper does not claim a closed class-membership theorem. Rather, it formulates an explicit conditional programme: if the five structural axioms (C1–C5) defining the Brazovskii class remain valid along the relevant RG flow, and if the five potential obstruction bounds (O1–O5) are confirmed by an unbiased spontaneous-emergence test on a sufficiently large lattice, then TECT becomes a candidate member of the Brazovskii universality class. We provide direct numerical estimates of (O1)–(O5) from the current branch-continuation data; these are conditional consistency checks, not closure of the class-membership question.

The structural significance for TECT is the following. *Once* such candidate class membership is itself promoted to a closed theorem by a future unbiased emergence test, mean-field Brazovskii predictions may be imported into TECT at the corresponding level of rigour, subject to the usual fluctuation and continuum-limit caveats that govern any application of universality classes to a microscopic candidate model. In particular, the freeze-out-moment identification of the correlation length $\xi \to a_{\mathrm{BCC}}$ (Paper 0) and the condensate density $\rho_{\mathrm{cond}}$ (Pillar 6) would, under such a verified class membership, inherit the Brazovskii critical-point geometry; the present paper does not by itself certify that import.

* jtkor@outlook.com

## II. SETTING AND MAIN RESULT

### A. Brazovskii free energy and the $(\Delta + q_0^2)^2$ structure

The Brazovskii free energy in its canonical form is

$$F[\Psi] = \int \left[ \frac{1}{2}|\nabla\Psi|^2 + \frac{\gamma}{2}(\Delta + q_0^2)^2|\Psi|^2 + \frac{\mu^2}{2}|\Psi|^2 + \frac{\lambda}{4}|\Psi|^4 \right] d^3x, \tag{1}$$

with order parameter $\Psi : \mathbb{R}^3 \to \mathbb{C}^n$, wavenumber $q_0 > 0$, and coupling constants $\gamma, \mu^2, \lambda$. The distinctive term $(\Delta + q_0^2)^2|\Psi|^2$ introduces a long-wavelength instability whose fastest-growing mode has wavenumber $|k| = q_0$.

### B. Five class-membership axioms

We define the Brazovskii universality class by five axioms:

*a. (C1) Finite wavenumber modulation —* The free energy possesses a finite-wavenumber instability at $|k| = q_0$ with $0 < q_0 < \infty$, distinct from the $|k| \to 0$ limit of the Standard Ising / XY classes.

*b. (C2) Non-local quartic gradient interaction —* The fourth-derivative term $(\Delta + q_0^2)^2$ dominates the structure near the critical point, so that the effective interaction range $\ell_{\text{eff}} \sim 1/q_0$ is comparable to the order-parameter correlation length $\xi$ at criticality.

*c. (C3) Amplitude-modulation landscape —* The free energy permits modulated structures with wavelength $2\pi/q_0$ and allows reentrant phase diagrams where the modulated phase may coexist with or precede uniform ordering, depending on the direction of approach to criticality.

*d. (C4) Real order-parameter support —* The order parameter is real or complex, without additional discrete symmetries (e.g., no $\mathbb{Z}_N$ pinning) that would change the lower critical dimension.

*e. (C5) Proximity to three dimensions —* The system exhibits critical behaviour in spatial dimensions $d \geq 2$, with the lower critical dimension $d_c = 1$. TECT operates at $d = 3$, in the mean-field regime where fluctuations are weakly relevant.

### C. Five obstruction bounds

Potential deviations from Brazovskii universality are ruled out by explicit estimates of five obstructions:

*a. (O1) Non-local term dominance —* The operator norm of the $(\Delta + q_0^2)^2$ term relative to the Laplacian over the critical-region wavenumber band $[q_0/2, 3q_0/2]$ satisfies $\|(\Delta + q_0^2)^2\|_{\rm op}/\|\Delta\|_{\rm op} > 5$ [3]. This ensures that mode $|k| = q_0$ is the dominant instability, not a subleading correction.

*b. (O2) Amplitude-field hierarchy —* The self-energy correction to the order-parameter field $\delta m^2$ due to the quartic coupling $\lambda$ satisfies $\delta m^2/|\mu^2| < 0.1$ in the critical region. This ensures that amplitude fluctuations do not drive the system into a different fixed point [4].

*c. (O3) Cubic invariance restoration —* At the critical point, the BCC lattice's discrete cubic symmetry restores to continuous $O(3)$ (or $O(n)$ for the TECT setting) to leading order, with $O(3)$-breaking corrections $< 0.05\%$ in the free-energy density. This rules out lock-in to other discrete structures (hexagonal, FCC) [5].

*d. (O4) Finite-size effects management —* For TECT runs on a lattice of size $N \times N \times N$ with $N = 32$ (typical in Pillar 6), the finite-size scaling $\Delta\epsilon_{\rm fs}/\epsilon_{\rm bulk} < 2\pi N^{-3}$ ensures that boundary effects do not mask the Brazovskii critical exponents. At $N = 32$, this bound yields $\Delta\epsilon_{\rm fs} < 10^{-5}$ in relative units [6].

*e. (O5) RG flow stability —* Along the critical trajectory in parameter space, the RG flow remains within 10% of the Brazovskii fixed point in the two-loop coupling $\beta$-function plane. Deviation beyond 10% would indicate crossover to a different universality class. Numerical harmonic-oscillator barrier-height integration confirms $|\Delta\beta_1| < 0.05$ [5].

**Conjecture 1** (Conditional Brazovskii-universality membership)**.** Under hypotheses (C1)–(C5) and provided that obstruction bounds (O1)–(O5) are verified by an unbiased spontaneous-emergence test on a sufficiently large lattice, the BCC topological condensate at the freeze-out transition governed by the free energy Eq. (1) *is conjectured to belong* to the Brazovskii universality class, in which case the critical exponents $\nu = 1/2$, $\alpha = 1/2$, and the scaling function $f(t, h) = t^{2-\alpha}\,\tilde{f}(h/t^{\Delta})$ would apply to TECT observables. The current evidence is branch-continuation numerical consistency with O1–O5, which constitutes conditional support but not a closed theorem; the unbiased emergence test and the beyond-two-loop RG-flow stability estimate remain open.

## III. CONSISTENCY CHECKS OF AXIOMS (C1)–(C5) AND OBSTRUCTION BOUNDS (O1)–(O5)

*Honest scope of this section.* The subsections below record *conditional consistency checks* of the axioms (C1)–(C5) and the obstruction bounds (O1)–(O5) at the present TECT operating point, on the basis of branch-continuation numerical estimates and analytical sketches. The phrasing "axiom verified" / "obstruction verified" below should be read as "consistent at the present canonical tier"; it is *not* a closure verdict and *not* a substitute for the unbiased spontaneous-emergence test of Conjecture 1.

### A. Axiom C1: Finite wavenumber modulation

The instability of the uniform phase $\Psi = 0$ is governed by the eigenvalues of the linearized operator $\delta^2 F/\delta\Psi\delta\Psi^*$ at $\Psi = 0$. In Fourier space, this operator acts as

$$\mathcal{L}_k = -|\nabla|^2 + \gamma(k^2 - q_0^2)^2 + \mu^2, \tag{2}$$

for mode $k$ with wavenumber $|k|$. The eigenvalue is $\lambda_k = -k^2 + \gamma(k^2 - q_0^2)^2 + \mu^2$. The most unstable mode satisfies $\partial\lambda_k/\partial k = 0$, which gives $|k| = q_0$ for the Brazovskii structure. At the TECT critical point, $\mu^2 < 0$ and $\gamma > 0$, so $\lambda_{q_0} = 0$ marks the onset of instability. Axiom C1 is verified.

### B. Axiom C2 and Obstruction O1

The relative dominance of the non-local term is assessed by comparing energy scales. Over the critical band $|k| \in [q_0/2, 3q_0/2]$, the Laplacian contributes energy $\sim k^2 \sim q_0^2$, while the $(\Delta + q_0^2)^2$ term contributes $\sim (k^2 - q_0^2)^2 \sim q_0^4$ for $k$ far from $q_0$ and $\sim 0$ at $k = q_0$. The Brazovskii mechanism relies on the interplay of these two scales. Obstruction O1 is verified via harmonic-oscillator barrier analysis in Math97-AddA, showing $\|(\Delta + q_0^2)^2\|_{\mathrm{op}} > 5 \times \|\Delta\|_{\mathrm{op}}$ in the critical region.

### C. Axiom C3 and Obstruction O2

Amplitude modulation is encoded in the nonlinear susceptibility $\chi^{(3)}(\omega, q_0; q_0, q_0)$, which couples three modes. In the Brazovskii class, this susceptibility is large but does not dominate the field equations. Obstruction O2 sets a quantitative bound: the self-energy shift $\delta m^2$ caused by the quartic interaction $\lambda|\Psi|^4$ must satisfy $\delta m^2/|\mu^2| < 0.1$. At the TECT working point ($\lambda \sim 1$, $|\mu^2| \sim 10^{-2}$ at freeze-out), this condition is satisfied with margin [4].

### D. Axiom C4

TECT's order parameter is a complex $n$-component field $\Psi \in \mathbb{C}^n$. This is standard for the Brazovskii framework and incurs no discrete symmetry constraints beyond the $O(n)$ rotational symmetry of the free energy. Axiom C4 is satisfied.

### E. Axiom C5 and Obstruction O3

The BCC lattice has cubic symmetry $O(h)$. At the critical point, the discrete cubic axes are isotropic directions of the order-parameter field $\Psi$. The O(n) continuum limit is achieved when the lattice spacing $a_{\mathrm{BCC}} \ll \xi$, permitting cubic anisotropy to fade into $O(n)$ symmetry. Obstruction O3 quantifies this fade: cubic anisotropy in the free-energy density contributes $< 0.05\%$ at $a_{\mathrm{BCC}}/\xi = 0.3$ (typical freeze-out condition) [5]. Axiom C5 is verified.

### F. Numerical closure: Obstructions O4 and O5

Obstructions O4 and O5 are verified numerically via the continuum-limit continuation protocol described in Math236. For representative runs at $N = 32, 48, 64$ (spatial lattice size), the finite-size scaling $\Delta\epsilon_{\mathrm{fs}}/\epsilon_{\mathrm{bulk}}$ extrapolates to $< 10^{-5}$ in the continuum limit (O4), and the RG-flow parameter $\beta_1$ remains within 5% of the Brazovskii fixed point (O5).

## IV. DISCUSSION AND OUTLOOK

The conditional Brazovskii-universality programme of this paper achieves two goals: (1) it provides a structured candidate classification of the TECT condensate's phase-transition mechanism along the canonical Brazovskii axes, and (2) it identifies the explicit obstruction bounds whose verification by an unbiased spontaneous-emergence test would promote the candidate class membership to a closed theorem; in turn, mean-field critical-exponent imports from the canonical Brazovskii literature would become admissible TECT predictions, subject to fluctuation and continuum-limit caveats.

The present result is conditional on five structural hypotheses (C1)–(C5) and on five obstruction bounds (O1)–(O5). The hypotheses are structural assertions about the form of the free energy (1); the bounds are *branch-continuation numerical estimates* (NOT analytic constant-bound theorems), which could in principle be violated by strong field-dependent renormalisation or unexpected higher-derivative corrections.

Should future numerical work uncover $\delta\beta_1 > 0.10$ or $\delta m^2/|\mu^2| > 0.1$ at the critical point, the universality-class assumption would require revision, and the critical exponents $\nu, \alpha$ would need to be re-derived from first principles. This forms a testable hypothesis: $\mathbf{H}_{\mathrm{Brazovskii}}$: obstruction bounds O1–O5 are all satisfied at the TECT freeze-out critical point.

A future verified establishment of Brazovskii-class membership would feed directly into the predictions of Paper 0 (cosmic origin of $\hbar$) via the Kibble–Zurek scenario, and into Paper 6 (Pillar 6 generations) via the freeze-out critical exponent $\nu = 1/2$ determining the bandwidth of the resultant field fluctuations. Until the universality-class membership is itself promoted to a closed theorem by the unbiased emergence test described above, both Paper 0 and Paper 6 carry the Brazovskii-class import as a programme hypothesis rather than as a guaranteed consequence; the alternative path of explicit derivation of critical exponents from microscopic first principles remains open as a far more computationally intensive programme.

## ACKNOWLEDGMENTS

This work is part of TECT, canonical archive at https://github.com/TECT-OpenPhysics/TECT.

[1] S. A. Brazovskii, Sov. Phys. JETP **41**, 85 (1975).
[2] J. Lee, TECT canonical repository, https://github.com/TECT-OpenPhysics/TECT (2026).
[3] J. Lee, TECT-Math97-Addendum-A, TECT internal note (2026).
[4] J. Lee, TECT-Math97-Addendum-B, TECT internal note (2026).
[5] J. Lee, TECT-Math97-Addendum-C, TECT internal note (2026).
[6] J. Lee, TECT-Math236: continuum-limit-scan numerical implementation, TECT internal note (2026).